SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA1707

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 4 **Total: Marks:40**

Annexure A

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Code of Conduct:

I J O Joanna Divine(192572003) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Joanna

INDUSTRY SCENARIO
A healthcare company has collected patient information such as age, blood pressure, cholesterol, glucose level, BMI, and family history. The objective is to develop an AI/ML system that can predict the risk of diabetes at an early stage and recommend personalised treatment actions such as **Diet, Exercise, Medication, and Monitor**.

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

1(a) Inductive Learning Approach

Inductive learning learns general patterns from known patient examples and uses those patterns to predict diabetes for new patients.

For this problem, a **supervised classification approach** is suitable because the dataset contains patient features and a known diabetes outcome.

Target Variable

The target variable is:

Diabetes = 0 or 1

- 0 → Non-diabetic
- 1 → Diabetic

Important Features

Feature	**Justification**
Glucose Level	High glucose is one of the strongest indicators of diabetes.
BMI	Higher BMI is associated with increased risk of insulin resistance and diabetes.
Age	Diabetes risk generally increases with age.
Blood Pressure	High blood pressure is commonly associated with metabolic disorders.
Family History	Genetic/familial factors can increase diabetes risk.
Cholesterol	Abnormal cholesterol levels can indicate metabolic risk.

The model learns the relationship between these features and the diabetes target and predicts the class for unseen patients.

1(b) Handling Class Imbalance

In a medical dataset, there may be many more non-diabetic patients than diabetic patients. This creates **class imbalance**.

For example:

- Non-diabetic = 900
- Diabetic = 100

A model trained directly on this dataset may predict most patients as non-diabetic and still obtain high accuracy while missing diabetic patients.

Recommended Strategy: SMOTE

I would use **SMOTE (Synthetic Minority Over-sampling Technique)**.

SMOTE creates synthetic examples of the minority class instead of simply duplicating existing diabetic records.

Advantages

1. Increases representation of diabetic patients.
2. Helps the classifier learn minority-class patterns.
3. Reduces the possibility of ignoring diabetic cases.
4. Usually preferable to random under-sampling when the dataset is not extremely large.

SMOTE should be applied **only to the training data**, after splitting the dataset, to avoid data leakage.

1(c) 70:30 Dataset Split

The dataset can be divided as:

- **70% Training data**
- **30% Testing data**

A **stratified split** should be used so that diabetic and non-diabetic cases maintain approximately the same proportion in both sets.

Justification

The 70:30 ratio provides enough data for model training while keeping a sufficiently large independent test set for reliable evaluation. This is particularly important in healthcare because the model must be tested on unseen patients.

Preprocessing

The following preprocessing steps are performed:

1. **Missing-value handling** – numerical missing values can be replaced using median imputation.
2. **Normalization/standardization** – numerical features such as glucose, BMI, age, and blood pressure can be standardized.
3. **Encoding** – categorical features such as family history can be converted into numerical values, for example:
   - No = 0
   - Yes = 1
4. **Outlier checking** – unrealistic medical values should be investigated.
5. **Class balancing** – SMOTE is applied only to the training set.

Impact

Preprocessing improves data quality and makes the features suitable for machine learning. Standardization is particularly important for models such as Logistic Regression.

TASK 2: DECISION TREE FOR DIAGNOSIS

2(a) Decision Tree Classifier

A **Decision Tree** classifies patients by repeatedly splitting the dataset according to feature values.

For example, the first three levels could be represented as:

```
                    Glucose
                   /       \
               Low           High
              /                 \
           BMI                 Age
          /   \               /   \
       Low     High        Young   Old
        |        |           |       |
   Non-Diabetic  Risk      Risk    Diabetic
```

The exact tree structure depends on the actual dataset.

Root Node Selection

The root node is selected using **Information Gain** or **Gini Index**.

Information Gain

Information Gain measures the reduction in uncertainty after splitting the data.

$$
IG = Entropy(parent) -
\sum \frac{N_i}{N}Entropy(child_i)
$$

The feature having the **highest Information Gain** is selected as the split.

Gini Index

$$
Gini = 1-\sum p_i^2
$$

The feature producing the **lowest weighted Gini Index** is preferred.

For a diabetes dataset, glucose level is expected to be a strong candidate for the root because it can provide substantial separation between diabetic and non-diabetic patients.

2(b) Model Evaluation

The Decision Tree should be evaluated using:

Accuracy

$$
Accuracy = \frac{TP+TN}{TP+TN+FP+FN}
$$

Precision

$$
Precision = \frac{TP}{TP+FP}
$$

Recall

$$
Recall = \frac{TP}{TP+FN}
$$

F1-Score

$$
F1 = 2\times\frac{Precision\times Recall}{Precision+Recall}
$$

A suitable result table can be presented as:

Metric	**Decision Tree**
Accuracy	90%
Precision	84%
Recall	88%
F1-Score	86%

Note: These values are example results. Actual values must be calculated using the supplied dataset.

False Negatives

A **False Negative (FN)** occurs when:

The patient actually has diabetes, but the model predicts non-diabetic.

In healthcare, false negatives are especially dangerous because a diabetic patient may not receive further testing or treatment.

Reducing False Negatives

To reduce false negatives:

- Increase the importance of the diabetic class.
- Use SMOTE during training.
- Adjust the classification threshold.
- Optimize for **recall/sensitivity** rather than accuracy alone.
- Perform additional clinical testing for high-risk patients.

High recall is important because missing a diabetic patient can delay diagnosis and treatment.

2(c) Post-Pruning

A fully grown Decision Tree can become too complex and overfit the training data.

Cost-complexity pruning removes unnecessary branches.

The objective can be represented as:

$$
R_\alpha(T)=R(T)+\alpha|T|
$$

where:

- $R(T)$ = model error
- $|T|$ = number of terminal nodes
- α = pruning parameter

Comparison

Model	Accuracy	Complexity	Overfitting
Pre-pruned Tree	Example: 88%	Low	Low
Post-pruned Tree	Example: 90%	Moderate	Lower

The exact values should be obtained experimentally.

Clinical Deployment

A **post-pruned tree** is generally preferable when it maintains comparable or better predictive performance because it is simpler, less prone to overfitting, and easier for healthcare professionals to interpret.

However, deployment should depend on validation performance, especially **recall, sensitivity, specificity, and calibration**, rather than accuracy alone.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

3(a) Logistic Regression

Logistic Regression can be used as the statistical learning model.

It predicts the probability of diabetes:

$$
P(Y=1)=\frac{1}{1+e^{-z}}
$$

where

$$
z=\beta_0+\beta_1X_1+\beta_2X_2+\cdots+\beta_nX_n
$$

The output is a probability between 0 and 1.

For example:

Patient probability of diabetes = 0.82

If the threshold is 0.5:

$0.82 > 0.5 \rightarrow$ Diabetic

Comparison

Metric	Decision Tree	Logistic Regression
Accuracy	Example: 90%	Example: 88%
AUC-ROC	Example: 0.91	Example: 0.93

Actual values must be calculated from the dataset.

When Logistic Regression is Preferable

Logistic Regression is useful when:

- The relationship between predictors and outcome is reasonably structured.
- Probability estimates are required.
- Interpretability is important.
- The organization needs to understand how individual variables affect risk.
- A simple baseline model is required.

In clinical environments, an interpretable statistical model can be valuable because healthcare professionals need to understand the reasoning behind risk predictions.

3(b) Feature Importance Analysis

Decision Tree

Feature importance can be calculated using the reduction in impurity produced by each feature.

A possible top-three ranking is:

1. **Glucose Level**
2. **BMI**
3. **Age**

Logistic Regression

For Logistic Regression, the magnitude of standardized coefficients can be examined.

A possible ranking is:

1. **Glucose Level**
2. **BMI**
3. **Family History**

Rank	Decision Tree	Logistic Regression
1	Glucose	Glucose
2	BMI	BMI
3	Age	Family History

The models may agree on the strongest predictors but differ on the third predictor.

Justification

Decision Trees identify features that produce the largest reduction in impurity, whereas Logistic Regression evaluates the contribution of variables through their coefficients. Therefore, differences in feature ranking are expected.

TASK 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

4(a) MDP Model

Treatment recommendation can be represented as a **Markov Decision Process (MDP)**.

An MDP consists of:

[
MDP=(S,A,P,R,\gamma)
]

States

Patient health can be represented using simplified states:

State	Description
S1	Low-risk / healthy
S2	Moderate diabetes risk
S3	High diabetes risk
S4	Diagnosed diabetes
S5	Improved/stable condition

The states represent the patient's health condition at different time points.

Actions

[
A={Diet, Exercise, Medication, Monitor}
]

- **Diet** → nutritional intervention.
- **Exercise** → physical activity recommendation.
- **Medication** → medication-related intervention under clinician supervision.
- **Monitor** → monitor glucose and other health indicators.

Reward Function

A simplified reward function can be:

Outcome	Reward
Significant health improvement	+10
Moderate improvement	+5
No change	0
Health deterioration	-10

The reward encourages the agent to select actions associated with improved patient outcomes.

Transition Dynamics

After an action, the patient's health state changes.

Example:

S3 -- Diet --> S2

S3 -- Exercise --> S2

S3 -- Monitor --> S3
S3 -- Medication --> S2
In a real healthcare system, these transitions must be learned from validated longitudinal clinical data and should not be invented by the AI agent.

4(b) Q-Learning
Q-Learning learns the expected future reward for each state-action pair.
The update equation is:
$$
Q(s,a)\leftarrow Q(s,a)+
\alpha[r+\gamma\max_{a'}Q(s',a')-Q(s,a)]
$$
where:

- (α) = learning rate
- (γ) = discount factor
- (r) = reward
- (s') = next state

Assume:
$$
\alpha=0.5,\quad \gamma=0.9
$$
Initially all Q-values are zero.
Episode 1 – Iteration 1
Suppose:
State = S3
Action = Diet
Reward = +5
Next State = S2
Since all initial Q-values are zero:
$$
Q(S3,Diet)=0+0.5[5+0.9(0)-0]
$$
$$
Q(S3,Diet)=2.5
$$
Episode 2 – Iteration 1
Suppose:
State = S2
Action = Exercise
Reward = +10
Next State = S5
$$
Q(S2,Exercise)
=0+0.5[10+0.9(0)-0]
$$
$$
Q(S2,Exercise)=5
$$
Episode 3 – Iteration 1
Suppose:
State = S3
Action = Medication

Reward = +5
Next State = S2
The maximum Q-value in S2 is currently 5.
[
$Q(S3,Medication)$
$=0+0.5[5+0.9(5)-0]$
]
[
$=0.5(9.5)$
]
[
$Q(S3,Medication)=4.75$
]

Second Iteration
Suppose the agent again reaches S3.
Update 1
For:
S3 → Diet → S2
Reward = 5
Current:
[
$Q(S3,Diet)=2.5$
]
Maximum Q-value at S2 = 5.
[
$Q(S3,Diet)=2.5+$
$0.5[5+0.9(5)-2.5]$
]
[
$Q(S3,Diet)=5.00$
]
Update 2
Suppose:
S2 → Exercise → S5
Reward = 10
Current:
[
$Q(S2,Exercise)=5$
]
[
$Q(S2,Exercise)$
$=5+0.5[10-5]$
]
[
$Q(S2,Exercise)=7.5$
]
Update 3
Suppose:
S3 → Medication → S2
Reward = 5
Current:

[
$$Q(S3,Medication)=4.75$$
]
Maximum Q-value at S2 = 7.5.
[
$$Q(S3,Medication) =4.75+0.5[5+0.9(7.5)-4.75]$$
]
[
$$=7.25$$
]
Therefore, after these updates:

State	Action	Q-value after Iteration 2
S3	Diet	5.00
S2	Exercise	7.50
S3	Medication	7.25

Five Patient Episodes

The agent can be trained for at least five episodes:

Episode	Initial State	Action	Next State	Reward
1	S3	Diet	S2	+5
2	S2	Exercise	S5	+10
3	S3	Medication	S2	+5
4	S2	Diet	S5	+5
5	S3	Monitor	S3	0

In an actual implementation, many more episodes would normally be required for reliable learning.

Convergence Criterion

Training can be considered converged when:
[
$$\max_{s,a}|Q_{new}(s,a)-Q_{old}(s,a)|<0.001$$
]
for a sufficiently large number of consecutive episodes.
This indicates that Q-values are no longer changing significantly.

4(c) Final Learned Policy

The policy selects the action having the highest Q-value:
[
$$\pi(s)=\arg\max_a Q(s,a)$$
]
Example:

State	Best Action	Reason
S1 – Low Risk	Monitor	Maintain healthy condition
S2 – Moderate Risk	Exercise	Encourages health improvement
S3 – High Risk	Medication*	Highest learned value in this example
S4 – Diagnosed	Monitor/Medication*	Requires clinical management
S5 – Improved	Diet	Maintain improvement

*Medication decisions must be made under qualified clinical supervision. An RL model should not independently prescribe or alter medication.

Ethical Considerations

1. Patient Safety

A reinforcement learning system should not directly experiment on patients. Unsafe actions must be restricted using clinical rules and human oversight.

2. Bias

If the training data is biased toward a particular demographic group, the agent may produce less accurate recommendations for underrepresented groups.

The system should therefore be evaluated across age, sex, ethnicity where legally and ethically appropriate, and other relevant patient subgroups.

3. Explainability

Healthcare professionals should be able to understand why a recommendation was generated. The system should provide the patient's relevant state, contributing factors, and supporting evidence.

4. Human-in-the-Loop

The final treatment decision should remain with a qualified healthcare professional. The AI should act as a **decision-support system**, not an autonomous doctor.

5. Privacy

Patient data must be protected using appropriate security, access control, anonymization/pseudonymization, and applicable healthcare data regulations.

6. Continuous Monitoring

After deployment, the model should be monitored for performance degradation, unexpected recommendations, bias, and safety issues.

Conclusion

The proposed system combines **inductive learning, Decision Trees, statistical learning, and Reinforcement Learning** to support diabetes diagnosis and treatment recommendation.

The Decision Tree provides an interpretable diagnostic model, while Logistic Regression provides probabilistic risk estimation. SMOTE helps address class imbalance, and a 70:30 stratified split provides separate training and testing data. Q-Learning can model sequential treatment decisions using patient states, actions, rewards, and transitions.

For real-world healthcare deployment, **patient safety, clinical validation, fairness, explainability, privacy, and human supervision are essential**. The system should therefore be used as a clinical decision-support tool rather than as an autonomous treatment authority.

RUBRICS FOR CALCULATION:

Criteria	Weightag e	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation